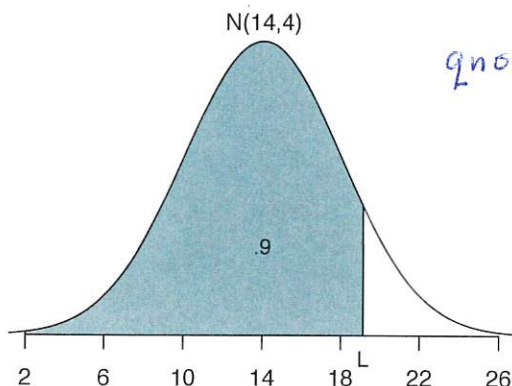


Worksheet: Normal Distributions Part II

1. Find the value L in the distribution $N(14, 4)$ that marks the 90th percentile. That is, find the value L such that $P(X < L) = 0.9$



$qnorm(0.9)$ = the z-score in $N(0,1)$ that has area 0.9 less than it, ~~so~~

$$\approx 1.28$$

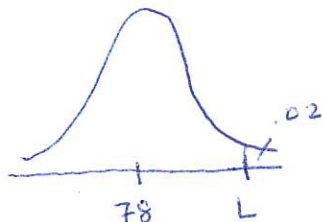
So we want L to have z-score 1.28

$$\frac{L - 14}{4} = 1.28$$

$$L - 14 = 4 \times 1.28$$

$$L = 14 + 4 \times 1.28 = \boxed{19.12}$$

2. Scores on a test are normally distributed with mean 78 and standard deviation 6. How high must you score to be in the top 2% of all scores?



We want L in $N(78, 6)$ with area .02 above it
so area .98 below it.

The z-score in $N(0,1)$ with area .98 below it is
 $z = qnorm(.98) = 2.054$

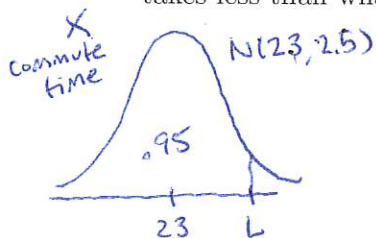
So solve for L :

$$\frac{L - 78}{6} = 2.054$$

$$\Rightarrow L = 78 + 6 \times 2.054$$

$$\boxed{L = 90.324}$$

3. The time it takes Ralph to commute to work has a normal distribution, with a mean of 23 minutes and a standard deviation of 2.5 minutes. On 95% of occasions the commute time takes less than what amount?



We want L in $N(23, 2.5)$ with area .95 below it

The z-score in $N(0,1)$ with area .95 below it is
 $z = qnorm(.95)$

$$z = 1.645$$

So solve for L :

$$\frac{L - 23}{2.5} = 1.645$$

$$\Rightarrow L = 23 + 2.5 \times 1.645$$

$$= \boxed{27.11 \text{ minutes}}$$